Unit: 6 Unit 4: Absolute Value
Topic: Solving Absolute Value Equations

Name: ____________
Date: ____________

Foundation (Jalapeno)

Q1. Solve $|x| = 5$
(A) $x = 5$
(B) $x = -5$
(C) $x = 5$ or $x = -5$
(D) $x = 0$
(E) $x = 10$

Q2. Solve $|2x| = 8$
(A) $x = 2$ or $x = -2$
(B) $x = 4$ or $x = -4$
(C) $x = 1$ or $x = -1$
(D) $x = 3$ or $x = -3$
(E) $x = 0$

Q3. Solve $|x - 3| = 2$
(A) $x = 1$ or $x = 5$
(B) $x = 3$ or $x = 7$
(C) $x = 2$ or $x = 4$
(D) $x = 0$ or $x = 6$
(E) $x = -1$ or $x = 9$

Q4. Solve $|x + 2| = 3$
(A) $x = -5$ or $x = 1$
(B) $x = -2$ or $x = 4$
(C) $x = 0$ or $x = 6$
(D) $x = -1$ or $x = 5$
(E) $x = 3$ or $x = 7$

Q5. Solve $|3x| = 12$
(A) $x = 2$ or $x = -2$
(B) $x = 4$ or $x = -4$
(C) $x = 3$ or $x = -3$
(D) $x = 1$ or $x = -1$
(E) $x = 0$

Q6. Solve $|x - 2| = 4$
(A) $x = 6$ or $x = -2$
(B) $x = 3$ or $x = 7$
(C) $x = 2$ or $x = 8$
(D) $x = 1$ or $x = 5$
(E) $x = 0$ or $x = 4$

Q7. Solve $|2x - 1| = 5$
(A) $x = 2$ or $x = 3$
(B) $x = 1$ or $x = 4$
(C) $x = 0$ or $x = 6$
(D) $x = 3$ or $x = 2$
(E) $x = -1$ or $x = 5$

Q8. Solve $|x + 1| = 2$
(A) $x = -3$ or $x = 1$
(B) $x = 0$ or $x = 4$
(C) $x = -2$ or $x = 2$
(D) $x = -1$ or $x = 3$
(E) $x = 2$ or $x = 6$

Open Ended Questions (FRQ)

Q9. Solve $|x - 4| = |2x + 1|$ and check for extraneous solutions

Q10. Solve $|3x - 2| = |x + 5|$ and check for extraneous solutions

Chef's Tips

- Check for extraneous solutions by plugging answers back into the equation
- Consider multiple cases when solving absolute value equations
- Always simplify your solutions to ensure they are in the simplest form possible